ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** O_2 and Fe	Oxygen and iron.
2	**(b)** sulphur	Sulphur is a non-metal; mercury is a metal that is liquid at room temperature.
3	**(d)** sonorous and hard	Alloy property.
4	**(a)** NH_3 and CH_4	Ammonia and methane.
5	**(c)** copper	Copper is an element; the others are mixtures of metals.
6	**(c)** a good conductor and ductile	Properties.
7	**(d)** compound	New substance.
8	**(b)** silicon	Metalloid features.
9	**(c)** good conductors of heat	Heat conduction.
10	**(c)** 1 : 8	2 g H : 16 g O.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	H and O are chemically combined in a fixed ratio (1); It has properties different from H2 and O2 (1).	1+1
12	C: 2, H: 6, O: 1 (1); Total 9 atoms (1).	1+1
13	Water has two hydrogen atoms and one oxygen atom (1); Formula is H2O (1).	1+1
14	Mixture retains properties; compound does not (1); Iron sulphide is not attracted by a magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Black mass forms (1); Magnet does not attract it (1); A compound is formed (1).	1+1+1
16	Lustrous (1); Malleable and ductile (1); Do not rust easily (1).	1+1+1
17	Bells: sonorous (1); Utensils: conduct heat (1); Wires: ductile and conduct current (1).	1+1+1
	OR: Magnet: attracts mixture only (1); Dilute acid: iron sulphide gives hydrogen sulphide smell (1); Mixture shows yellow and grey particles (1).	1+1+1
18	Chemically inert (1); Helium in balloons (1); Neon in advertisement lights (1).	1+1+1
	OR: X is a metal (1); Malleable (1); Sonorous (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mixture: magnet separates iron (1); Mixture retains yellow and grey colours (1); After heating: black non-magnetic iron sulphide (1); New properties (1); Compound formed by chemical change (1).	1+1+1+1+1
	OR: Iron in blood (1); Calcium in bones (1); Oxygen to breathe (1); Copper wires (1); Carbon fuels (1).	1+1+1+1+1
20	Element: one kind of atom (1); Compound: chemically combined (1); Mixture: physically combined (1); Separation methods (1); Examples for each (1).	1+1+1+1+1
	OR: Mercury and oxygen formed (1); Compound decomposed on heating (1); Gas is oxygen (rekindles splinter) (1); Compounds break into elements chemically (1); Mercury is a liquid metal (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.
- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** graphite (carbon)	Exception.
2	**(a)** malleable and non-toxic	Thin sheets.
3	**(d)** a homogeneous mixture	Solution.
4	**(b)** 7	2 + 1 + 4.
5	**(a)** reaction with sulphur compounds	Black layer.
6	**(d)** iron and sulphur mixture	Iron is magnetic.
7	**(c)** 4	Na, H, C, O.
8	**(b)** stainless steel	Alloy with chromium.
9	**(c)** baking soda	Sodium bicarbonate.
10	**(d)** 24	6 + 12 + 6.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Mixture: water, fat, proteins, sugar (1); Composition varies and components can be separated (1).	1+1
12	Good conductors (1); Ductile and cost-effective (1).	1+1
13	They have free electrons (1); Electrons carry energy and charge quickly (1).	1+1
14	Sonorous property of metals (1); Bells and musical instruments (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Elements: copper, oxygen (1); Compounds: baking soda, ammonia (1); Mixtures: sea water, brass (1).	1+1+1
16	Pass current through acidified water (1); Hydrogen and oxygen formed in 2 : 1 by volume (1); Water breaks into elements (1).	1+1+1
17	Diamond: hardest natural substance (1); Graphite: soft and conducting (1); Same element, different forms (1).	1+1+1
	OR: Brass: mixture of metals with varied composition (1); Water: fixed composition H2O (1); Alloy components keep metallic properties (1).	1+1+1
18	Sugar solution has uniform composition (1); Muddy water has visible particles (1); Particles settle in muddy water (1).	1+1+1
	OR: Copper for wire (1); Plastic for insulation (1); Aluminium for foil (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lustre, malleability, ductility (1); Conductivity, sonority (1); Non-metals opposite (1); Mercury liquid, graphite conducts (1); Diamond hard (1).	1+1+1+1+1
	OR: Dissolving salt and sugar (1); Filtering tea and juice (1); Separating cream (1); Sedimentation/filtration of water (1); Use of physical methods (1).	1+1+1+1+1
20	Magnet for iron (1); Sublimation for camphor (1); Add water, filter to remove sand (1); Evaporate for salt (1); Physical methods only (1).	1+1+1+1+1
	OR: Noble gases: inert (1); Helium, neon, argon uses (1); Chlorine: reactive, disinfectant (1); Do not form many compounds (1); Use in lamps/purification (1).	1+1+1+1+1

Mistakes to avoid:

- Calling a molecule such as O_2 a compound; a compound needs *different* elements.
- Saying air or an alloy is a compound; they are mixtures.
- Assuming all metals are hard and solid; mercury is a liquid and sodium is soft.
- Writing wrong symbols (S for sodium, P for potassium); use Na and K.
- Using a physical method (filtration, magnet) to separate the elements of a compound.

- Writing that all non-metals are poor conductors; graphite is an exception.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

THE MASTERMIND COACHING CLASSES

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Nature of Matter: Elements, Compounds and Mixtures** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** chemically inert	Noble gases.
2	**(c)** water	Water is a compound.
3	**(a)** P_4	Phosphorus molecule.
4	**(c)** reacts vigorously with air and water	Reactivity.
5	**(a)** carbon dioxide	Carbon burns to CO2.
6	**(c)** water	Compound.
7	**(b)** evaporation and condensation	Physical methods.
8	**(d)** separating funnel	Immiscible liquids.
9	**(b)** dissolve in water, filter, evaporate	Sugar dissolves.
10	**(c)** carbon	Allotropes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Components are chemically bonded (1); Physical methods do not break bonds (1).	1+1
12	Better strength and resistance to rust (1); Steel is stronger than iron (1).	1+1
13	Element molecule: same atoms, e.g. O_2 (1); Compound molecule: different atoms, e.g. H_2O (1).	1+1
14	Sodium (1); Chlorine in salt; carbon in baking soda (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	H2SO4: 7 atoms; H, S, O (1); NH3: 4 atoms; N, H (1); CH4: 5 atoms; C, H (1).	1+1+1
16	Salt and sand: dissolve in water, filter, evaporate (1); Iron and sulphur: use magnet (1); Both are physical methods (1).	1+1+1
17	Components keep properties (1); Oxygen supports burning; nitrogen does not (1); Composition varies (1).	1+1+1
	OR: Chlorine: purifying water (1); Sulphur: matches, medicines (1); Nitrogen: fertilisers and packaging (1).	1+1+1
18	Metals conduct well; metalloids partly (1); Metals are malleable; metalloids brittle (1); Iron vs silicon (1).	1+1+1
	OR: Copper and tin are melted together (1); Mixture is cast into moulds (1); Hard, sonorous alloy forms (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steel: Fe + C, strong (1); Stainless steel: Fe + Cr + Ni, rust-proof (1); Brass: Cu + Zn, fittings (1); Bronze: Cu + Sn, bells (1); Improved properties (1).	1+1+1+1+1
	OR: All made of carbon atoms only (1); Diamond: hard, non-conductor (1); Graphite: soft, conductor (1); Charcoal: porous, fuel (1); Different arrangement of atoms (1).	1+1+1+1+1
20	H atom: 1 (1); H2: 2, O2: 2 (1); H2O: 3 (1); CO2: 3 (1); NH3: 4 (1).	1+1+1+1+1
	OR: (a) Filtration (1); (b) Evaporation or chromatography (1); (c) Separating funnel (1); (d) Sublimation (1); (e) Hand picking or sieving (1).	1+1+1+1+1

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